Enterprise Google Cloud Data Platform Blueprint

Volume 04: Data Engineering

Pipeline patterns, reusable components and engineering standards

Document attribute	Value
Version	0.1 reference implementation
Date	2026-07-10
Intended audience	Executive, architecture, security, platform engineering, data, AI, SRE and FinOps stakeholders
Scope	Fortune 100 reference blueprint for petabyte-scale Google Cloud data platform
Status	Generated implementation starter set; verify service-specific settings against current Google Cloud documentation

Table of Contents

1. 1. Engineering Strategy
2. 2. Pipeline Standards
3. 3. Reusable Templates
4. 4. Operations
5. 5. Pitfalls

Inherited Fortune 100 Assumptions

Area	Assumption
Scale	10-50 PB historical data; 50-250 TB/day new data; billions of events/day
Organization	Global enterprise with central platform team and federated domain teams
Regulatory	GDPR, HIPAA, PCI DSS, SOC 2, ISO 27001, NIST and regional residency constraints may apply
Availability	Tier 0 RTO <1 hour and RPO <5 minutes; Tier 1 RTO <4 hours and RPO <15 minutes
Cloud posture	Hybrid and multi-cloud integration with Google Cloud as the enterprise data platform foundation
Security	Zero Trust, least privilege, private connectivity, service perimeters, CMEK for restricted data

1. Engineering Strategy

Standardize pipeline templates for batch, micro-batch, CDC and streaming. Reusable components reduce delivery time and improve reliability.

2. Pipeline Standards

Every production pipeline must implement configuration-as-data, schema validation, DQ checks, idempotency, retries, DLQ/quarantine, observability, SLOs and rollback procedures.

Pattern	Primary Service	Use Case
Batch ETL	Dataflow/Beam	Large-scale file parsing and normalization
ELT	Dataform/BigQuery SQL	Warehouse transformations and data marts
Streaming	Pub/Sub + Dataflow	Low-latency events and CDC enrichment
Spark	Dataproc/Serverless Spark	Existing Spark migrations and advanced transformations
Orchestration	Cloud Composer	Complex DAG dependencies

Figure: Data Engineering Pipeline Patterns

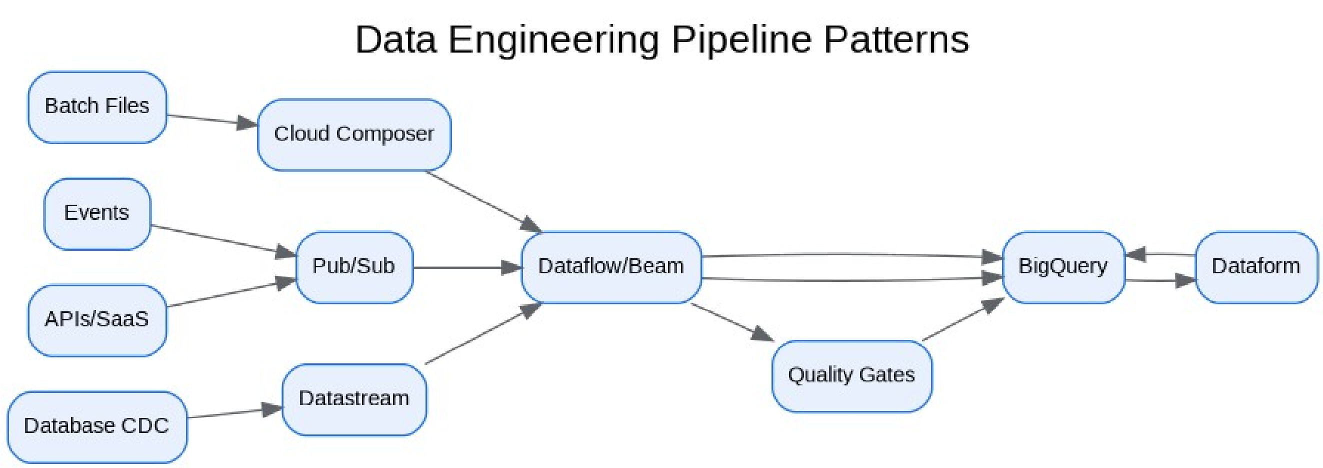
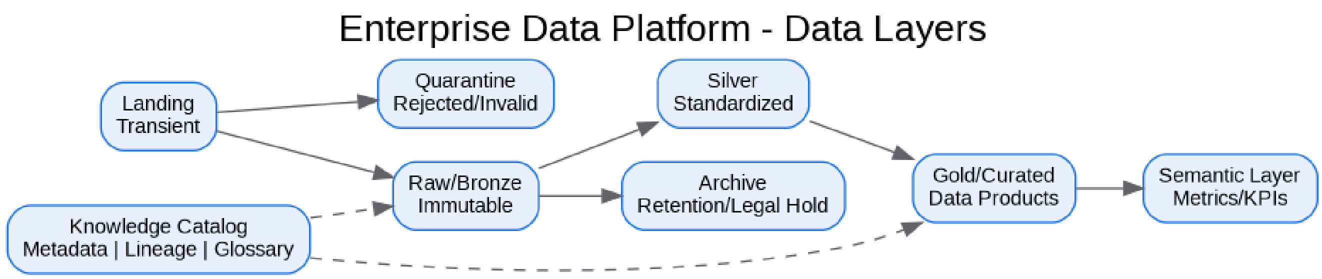


Figure: Enterprise Data Platform - Data Layers



3. Reusable Templates

Reference implementations are provided for Airflow DAGs, Dataflow Beam jobs, PySpark, Dataform SQLX and BigQuery SQL.

4. Operations

Pipelines emit freshness, volume, error-rate, backlog, cost and data-quality metrics. SLO violations trigger alerts and domain-owner escalation.

5. Pitfalls

- Building one-off pipelines without templates.
- Using streaming when business latency is batch.
- Missing reconciliation and replay design.
- Mixing security credentials across environments.

Architecture Decision Records

ADR	Decision	Problem	Alternatives	Decision	Tradeoffs/Risks	Mitigation
ADR-001	Google Cloud as enterprise data platform	Fragmented analytics estate	Continue legacy; multi-cloud split; Google Cloud data platform	Adopt Google Cloud data platform with hybrid integration	Requires operating-model change	Phased landing zone, domain pilots, and governance automation
ADR-002	BigQuery analytical core	Need PB-scale SQL analytics	Legacy EDW; Spark-only; third-party warehouse	Use BigQuery for curated/serving layers	Cost governance required	Reservations, labels, query controls, FinOps dashboards
ADR-003	Lakehouse with Cloud Storage and BigLake	Need open data and governed access	BigQuery-only; unmanaged object lake	Use Cloud Storage + BigLake + BigQuery	Open table governance maturity required	Catalog, policies, certified data-product lifecycle
ADR-004	Zero Trust and VPC SC	Exfiltration risk and regulated data	Network-only perimeter; IAM-only controls	Use identity, network, data and service perimeters	Service exceptions require testing	Perimeter dry runs and explicit ingress/egress policies

Risks and Mitigations

Risk	Impact	Mitigation
Underestimated legacy complexity	High	Complete source, pipeline, report and

		dependency assessment before migration waves
Data ownership gaps	High	Create domain owner RACI and data-product certification gate
Cost overrun	High	Enforce labels, budgets, reservations, query guardrails and monthly FinOps reviews
Security control drift	Critical	Deploy IaC, policy-as-code, automated IAM review and security posture monitoring
AI data leakage	Critical	Use certified datasets, service perimeters, DLP, RAG guardrails and model governance

Production Readiness Checklist

- All projects, networks, KMS keys and service accounts deployed through Terraform.
- All data products registered with owner, steward, classification, SLO, contract and support model.
- All restricted paths validated against VPC Service Controls and private connectivity requirements.
- All pipelines implement retries, dead-letter handling, reconciliation and operational metrics.
- All dashboards, alerts, runbooks and escalation routes tested before production cutover.
- All cost allocation labels present and billing export dashboards available to product owners.

References

- Google Cloud Architecture Framework: https://docs.cloud.google.com/architecture/framework
- Google Cloud Enterprise Foundations Blueprint: https://docs.cloud.google.com/architecture/blueprints/security-foundations
- Google Cloud Terraform blueprints and modules: https://docs.cloud.google.com/docs/terraform/blueprints/terraform-blueprints
- Build data products in a data mesh: https://docs.cloud.google.com/architecture/build-data-products-data-mesh
- Architecture and functions in a data mesh: https://docs.cloud.google.com/architecture/data-mesh
- Knowledge Catalog formerly Dataplex: https://cloud.google.com/products/knowledge-catalog
- Optimize costs with FinOps hub: https://docs.cloud.google.com/billing/docs/how-to/finops-hub
- VPC Service Controls supported products: https://docs.cloud.google.com/vpc-service-controls/docs/supported-products